Government of Karnataka

GOVERNMENT POLYTECHNIC
DEPARTMENT OF TECHNICAL EDUCATION KARNATAKA

Department of Collegiate & Technical Education

SRI JAYACHAMARAJENDRA GOVERNMENT POLYTECHNIC

BENGALURU - 560001

2025-26

A Report on

"INTERNSHIP"

Submitted in partial fulfilment for the Award of Diploma in

COMPUTER SCIENCE AND ENGINEERING

by

Megha Bharathi M
(102CS23031)

Under the Supervision of
Mr. Santosh Jena
NetLabs Global IT Services Pvt Ltd, Bengaluru

Under the Guidance of
Mr. Raghavendra

Government of Karnataka

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CERTIFICATE

This is to certify that **Report** submitted by **Megha Bharathi M**, (**102CS23031**) represents his work during the **2025-26** academic year at **NetLabs Global IT Services Pvt Ltd, Bengaluru**, under the Specialization Pathway **"INTERNSHIP(AI&ML)-20CS61S"** fulfilling the requirements for the award of the Diploma in Computer Science and Engineering.

Cohort Owner	Program Coordinator	Principal
Mr Raghavendra M Y	Mrs. Haritha S M	Sri Harish

Examiners:

1._____ 2. _____

INTERNSHIP REPORT

NetLabs Global IT Services Pvt Ltd

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to our respected **Mr Harish M** Pricipal for providing a supportive and motivating environment that encourages learning and professional growth.

I extend my sincere appreciation to **Mrs. Haritha** Head of the Department of Computer Science Engineering for their valuable guidance, encouragement, and continuous support.

I would like to express my sincere thanks to **Mr. Kishore** Placement Officer for providing me with the opportunity to undergo this internship and gain practical exposure to industry-level work. This opportunity has greatly contributed to my learning and professional development.

I am deeply grateful to the **Mr. Raghavendra** Cohort owner for his constant support, guidance, and encouragement throughout the internship period. His valuable suggestions and mentorship have been instrumental in helping me successfully complete my project.

I would like to express my deepest appreciation to **Mr. Santosh Jena** Industry Supervisor whose expertise and guidance have been invaluable in providing direction to my work. His encouragement and insightful feedback helped me overcome challenges and achieve the desired outcomes in my project titled "Agriculture Drought Prediction."

Finally, I would like to express my deepest gratitude to my **parents** for their unconditional support, constant motivation, and belief in me. I also extend my heartfelt thanks to my **friends** for their continuous support, cooperation, and encouragement.

Megha Bharathi M

102CS23031

EXECUTIVE SUMMARY

In today's data-driven world, leveraging technology to solve real-world problems has become essential across various domains. This project work focuses on two key applications: **Agriculture Drought Prediction and GP-SMS Data Automation**, both aimed at improving decision-making and operational efficiency using data analysis and automation techniques.

Agriculture Drought Prediction addresses the critical problem of unpredictable drought conditions, which can severely impact agricultural productivity and resource planning. Farmers and authorities often lack timely insights to take preventive measures.

To solve this, a machine learning-based system is developed that predicts the likelihood of drought using environmental parameters such as **rainfall, temperature, humidity, and soil moisture.** The system involves data preprocessing, feature selection, and model training using classification algorithms like Logistic Regression and Decision Trees.

A user-friendly web application is built using **Flask**, allowing users to input data and receive real-time predictions. This solution supports early warning, better planning, and efficient water resource management.

In addition to this, the GP-SMS Automation project focuses on improving the management of patient records by automating data processing workflows. The problem addressed here is the manual handling and transfer of patient-related data from multiple sources, which is time-consuming and prone to errors.

The solution involves building an automation pipeline using Python and data processing techniques. Two raw input datasets—**GP Master data and SMS Backup data**—are integrated and processed.

The system uses parameters such as last processed time to filter and update records efficiently. This automation ensures accurate data transfer from one source to another, reduces manual effort, and enhances data reliability.

Both projects demonstrate the effective use of machine learning and automation technologies in solving practical problems.

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Abbreviations

SHORT-FORM	FULL-FORM
GP	General Practitioner
RPA	Robotic Process Automation
IT	Information Technology
AI	Artificial Intelligence
CEO	Chief Executive Officer
HTTP	Hyper Text Transfer Protocol
API	Application Programming Interface
REST APIs	Representational State Transfer Application Programming Interface
HRMS	Human Resource Management System
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
SMS	Short Message Service

CHAPTER-01

COMPANY PROFILE

1.1 Overview Of the Organisation

Fig 1.1 Company logo

Netlabs Global IT Services Pvt. Ltd. is a leading global IT solutions and services company that provides end-to-end technology and business transformation solutions. The organization focuses on helping businesses improve efficiency, reduce operational costs, and achieve digital transformation through innovative technologies.

The company offers a wide range of services including IT infrastructure management, cloud computing, automation, artificial intelligence (AI), and Robotic Process Automation (RPA). It serves clients across various industries by delivering scalable, secure, and high-performance solutions.

Netlabs Global also emphasizes talent development and provides opportunities for interns and professionals to work on real-time projects, enabling them to gain practical industry experience.

1.2 Vision and mission of the Organisation

1.2.1 *Vision* :-

"To create endless possibilities of growth and well-being for all our stakeholders – customers, employees, partners, investors and society."

The vision of Netlabs Global IT Services Pvt. Ltd. is to create growth opportunities and deliver innovative technology solutions that benefit customers, employees, and society. The company aims to be a trusted global leader in digital transformation.

1.2.2 *Mission* :-

"We strive to be a Global Business Transformation enabler by helping our customers adopt smart cutting-edge technologies, smart industry practices and smart people talent solutions to stay ahead of the curve."

The mission is to help organizations improve their business operations by using advanced technologies, skilled professionals, and efficient processes, while ensuring quality service and customer satisfaction.

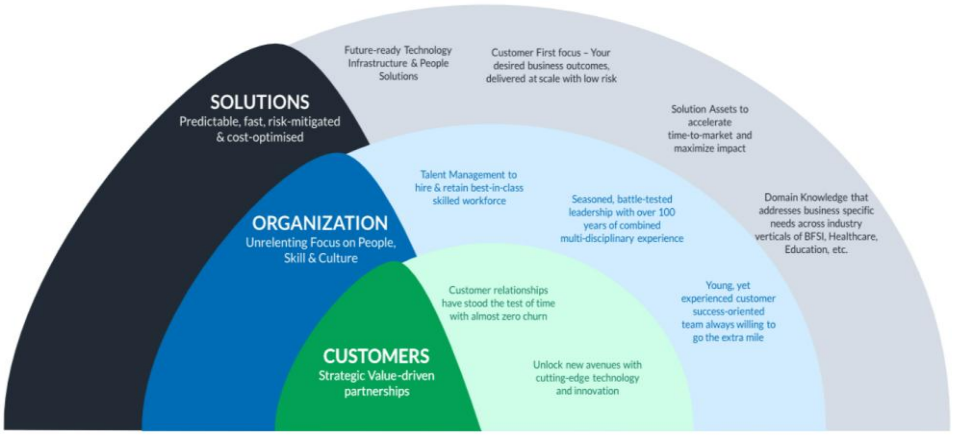


Fig 1.2 Value Proposition

1.3 Organisation Structure :-

Fig 1.3 Organisation Structure

The organization follows a well-defined hierarchical structure to ensure smooth functioning and efficient management. The top management, including the CEO and other senior executives, is responsible for strategic planning and decision-making.

Below them, team leads manage day-to-day project activities, assign tasks, and monitor progress. Developers, engineers, and interns work under their guidance to execute tasks effectively. This structure helps in maintaining clear communication, proper coordination, and timely delivery of projects.

1.4 Roles and Responsibilities :-

- **CEO / Managing Director:**

Oversees overall business operations, strategic decisions, and company growth.

- **Chief Strategy Officer:**

Plans business strategies and ensures long-term development and innovation.

- **Chief Financial Officer:**

Manages financial planning, budgeting, and company resources.

- **Team Lead (Seema):**

Supervises project execution, assigns tasks, and ensures timely delivery.

- **Developers / Engineers:**

Design, develop, and maintain software solutions and systems.

- **Interns / Trainees:**

Assist in project tasks, learn new technologies, and support development activities.

1.5 Products and Services :-

Netlabs Global offers a wide range of services, including:

- Managed IT Services (Infrastructure, Cloud, Applications)
- AI & RPA (Automation Solutions)
- Cyber Security & Compliance Services

Fig 1.4 Cyber Resilience and compliance

- Data Center and Cloud Services

Fig 1.5 Data Center Solutions

- Talent-as-a-Service (Workforce Solutions)

Netlabs Global IT Services Pvt. Ltd. offers a range of IT solutions including managed IT services, cloud computing, and data center management. The company also provides advanced solutions in Artificial Intelligence (AI) and Robotic Process Automation (RPA) to automate business processes. In addition, it offers cybersecurity services to ensure data protection and software development services to build efficient and scalable applications. These services help organizations improve productivity, security, and overall performance.

1.6 Conclusion :-

Netlabs Global IT Services Pvt. Ltd. is a growing IT organization that focuses on delivering innovative and reliable technology solutions. The company provides a strong platform for learning and development through real-time projects and industry exposure. Its structured workflow, advanced services, and focus on digital transformation make it an ideal organization for gaining practical knowledge and professional experience.

CHAPTER-02

OJT-1 ROLES AND RESPONSIBILITIES

2.1 Overview Of the Work done :-

During OJT-1, I worked on the **GP-SMS Automation Project**, which focuses on automating the process of handling SMS data and generating structured Excel reports. The main objective of this project was to reduce manual effort and improve accuracy in data processing through automation.

Along with project work, I learned Python programming concepts such as iterables, functions, and data handling techniques. I also gained hands-on experience in API development, including **REST APIs and HTTP methods like GET, POST, PUT, and DELETE.**

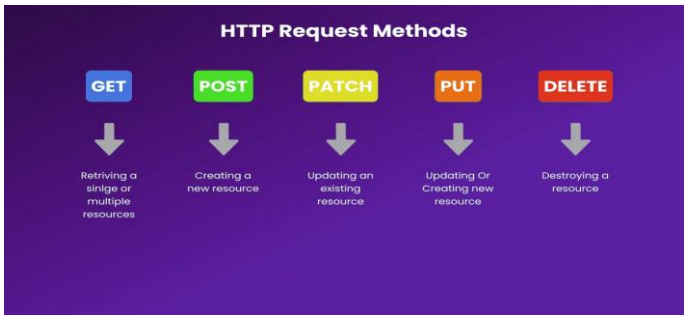


Fig 2.1 HTTP METHODS

I developed multiple APIs such as authentication and authorization APIs using different methods like token-based, session-based, and API key authentication. In addition, I worked on real-time use cases like **HRMS API and banking API**, which helped me understand how APIs are used in practical applications.

This phase provided strong foundational knowledge in automation, API development, and data processing, and helped me apply theoretical concepts to real-world scenarios.

2.2 Roles and Responsibilities :-

During OJT-1, I gained both fundamental and advanced technical knowledge in programming, API development, and automation.

I learned **Python programming basics** including variables, loops, functions, and iterables, which helped in building a strong foundation for automation tasks. I also worked on **file handling and data processing**, using libraries like Pandas to clean, transform, and organize data efficiently.

In addition, I gained in-depth knowledge of **API development**, especially REST APIs. I learned how APIs work in real-world applications and how to use different **HTTP methods** such as GET, POST, PUT, and DELETE for data communication between systems.

A major part of my learning involved **authentication and authorization mechanisms**, where I implemented:

- Token-based authentication
- Session-based authentication
- API key authentication

I also developed multiple real-time APIs such as **HRMS API, Banking API, and Invoice-related APIs**, which helped me understand practical implementation and system integration.

Furthermore, I was introduced to **automation tools like Automation Anywhere A360**, which enhanced my understanding of Robotic Process Automation (RPA) and how automation can be applied to reduce manual effort.

Overall, this phase helped me build strong technical skills in **Python, APIs, data handling, and automation**, preparing me for real-world industry applications.

✅ This is:

✓ Detailed (good for marks)

✓ Covers ALL your learning

✓ Sounds professional 💯

- If you want next
  I can make **same detailed section for OJT-2 (even stronger)**

2.3 Skills and Technologies Learned :-

During this phase, I learned both programming fundamentals and advanced concepts:

- **Python Basics:** Variables, loops, functions, and iterables
- **Data Handling:** File operations and data processing using Pandas
- **API Development:**
  - REST API concepts
  - HTTP methods (GET, POST, PUT, DELETE)
- **Authentication Methods:**
  - Token-based authentication
  - Session-based authentication
  - API key authentication
- Built multiple APIs such as:
  - HRMS API
  - Banking API
  - Authentication & Authorization APIs

2.4 Application of Technical Knowledge :-

The knowledge gained during OJT-1 was applied to real-time tasks in the GP-SMS automation project. Python programming concepts were used to develop scripts for data extraction, cleaning, and transformation. APIs were utilized to fetch and transfer data efficiently between systems, enabling automation of repetitive tasks.

Fig 2.2 GP-SMS(KDAH Hospital)

The understanding of REST APIs and HTTP methods helped in building and testing multiple APIs for different use cases. Authentication and authorization techniques were applied to ensure secure data access and handling. Additionally, data processing skills using Pandas and Excel were used to organize and generate structured reports.

Overall, the combination of programming, API development, and automation concepts was effectively applied to create efficient and reliable solutions.

2.5 Value Addition to Organization :-

The work carried out during OJT-1 added significant value by improving the efficiency and accuracy of data processing tasks. Automation reduced manual effort and minimized errors in handling large datasets.

The development of APIs provided a scalable and reusable approach for data communication, making the system more flexible and efficient. The project also helped in faster report generation and better data management.

This phase enhanced my problem-solving skills, technical knowledge, and ability to work on real-time industry projects, contributing to both personal growth and organizational productivity.

2.6 Workflow / Process Followed :-

- The GP-SMS automation project followed a structured workflow to ensure efficient data processing and accurate report generation.
- Initially, SMS data was **downloaded from the source** using Python scripts and APIs. The raw data often contained inconsistencies and unstructured information, so the next step involved **data cleaning and preprocessing**, where unwanted characters, duplicates, and invalid entries were removed.
- After cleaning, the data was **transformed into a structured format** and converted into Excel files for better readability and processing. Important fields such as phone numbers and timestamps were formatted properly, including splitting date and time values where required.
- The processed data was then **merged with the GP master dataset** to map relevant information and enrich the dataset. Record matching techniques were applied to ensure correct mapping and avoid duplication.
- Further, unique identifiers were generated for each SMS record to maintain data consistency. Finally, the system produced a **final automated Excel report**, which contained clean, structured, and accurate data ready for analysis.
- This workflow ensured smooth execution of tasks, reduced manual intervention, and improved overall efficiency of the data processing system.

Fig 2.3 GP-SMS Workflow

2.7 Conclusion :-

The OJT-1 phase provided a strong foundation in Python programming, API development, and automation. It helped in understanding how theoretical concepts can be applied to real-world problems. The hands-on experience gained through the GP-SMS project improved technical skills, increased confidence, and prepared me for advanced tasks in the next phase of the internship.

CHAPTER -03

OJT-2 ROLES AND RESPONSIBILITIES

3.1 Overview of Work Done :-

During OJT-2, I worked on **Invoice Management Systems and financial data processing**. The training focused on understanding the concept of invoices, their importance in business transactions, and how they are used to track payments and maintain financial records.

Along with theoretical understanding, I developed multiple APIs related to invoice management. I also worked on extracting invoice data into Excel files and performing operations like data merging, record matching, and validation. This phase provided practical exposure to handling real-time financial data and building scalable systems.

3.2 Roles and Responsibilities :-

- Developed APIs for invoice management
- Worked on extracting invoice data into Excel format
- Performed data merging and record matching across multiple Excel files
- Processed and validated financial data
- Assisted in building structured workflows for invoice handling

3.3 Skills and Technologies Learned :-

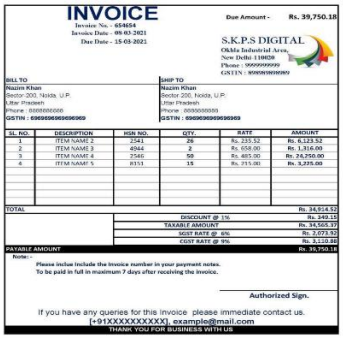
3.3.1 Invoice Concepts

- Understanding invoices and their importance in business transactions
- Purpose of invoices in tracking payments and financial records

3.3.2 API Development

Built multiple APIs related to invoice management:

- Invoice API



INVOICE

Invoice No. - 654654
Invoice Date - 05-03-2021
Due Date - 15-03-2021

Due Amount - Rs. 39,750.18

S.K.P.S DIGITAL
Okhla Industrial Area,
New Delhi-110020
Phone - 9999999999
GSTIN - 999999999999

BILL TO	SHIP TO
Nazim Khan Sector 200, Noida, U.P. Uttar Pradesh Phone : 8888888888 GSTIN : 696969696969696969	Nazim Khan Sector 200, Noida, U.P. Uttar Pradesh Phone : 8888888888 GSTIN : 696969696969696969

SL NO.	DESCRIPTION	HSN NO.	QTY.	RATE	AMOUNT
1	ITEM NAME 2	2541	26	Rs. 235.52	Rs. 6,123.52
2	ITEM NAME 3	4944	2	Rs. 658.00	Rs. 1,316.00
3	ITEM NAME 4	3546	50	Rs. 485.00	Rs. 24,250.00
4	ITEM NAME 5	8113	15	Rs. 215.00	Rs. 3,225.00
TOTAL					Rs. 34,914.52
			DISCOUNT @ 1%		Rs. 349.15
			TAXABLE AMOUNT		Rs. 34,565.37
			SGST RATE @ 6%		Rs. 2,073.92
			CGST RATE @ 6%		Rs. 2,120.89
PAYABLE AMOUNT					Rs. 39,750.18

Note:-
Please inclue include the invoice number in your payment notes.
To be paid in full in maximum 7 days after receiving the invoice.

Authorized Sign.

If you have any queries for this invoice please immediate contact us.
[+91XXXXXXXXXX], example@mail.com

THANK YOU FOR BUSINESS WITH US

Fig 3.1 Invoice Receipt

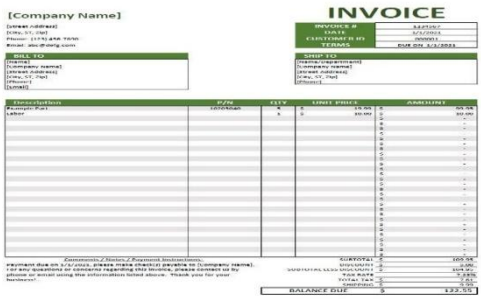
- Customer Balance API
- Invoice Payment API
- Credit Memo API
- Aging Report API
- Ledger API

3.3.3 Technical Concepts

- REST API development
- Handling financial data through APIs
- Data extraction and transformation

3.3.4 Excel Operations

- Extracting invoice data into Excel



[Company Name]

INVOICE

Fig 3.2 Invoice Template (Excel)

- Merging data from multiple Excel sheets
- Matching records between datasets

- Data cleaning and validation

3.4 Application of Technical Knowledge :-

The knowledge gained during training was applied to develop real-time invoice management solutions. API development skills were used to build systems that manage invoices, payments, and financial reports.

Understanding of REST APIs and HTTP methods helped in creating efficient and scalable systems. Data processing skills were applied to extract, merge, and validate large datasets.

Excel operations and record matching techniques ensured data accuracy and consistency. Overall, technical knowledge was effectively applied to solve real-world business problems.

3.5 Value Addition to Organization :-

The work carried out during OJT-2 improved the efficiency of invoice processing and reduced manual effort in handling financial data. Automation of data extraction, merging, and validation helped in minimizing errors and saving time.

The APIs developed provided a structured and scalable solution for managing invoices and financial records. This improved system efficiency and data accessibility.

This phase also enhanced my understanding of business processes and strengthened my technical and analytical skills.

3.6 Workflow / Process Followed :-

The workflow in OJT-2 began with understanding invoice structures and business requirements. Based on this, APIs were designed and developed to handle different invoice operations such as creation, payment tracking, and reporting.

Invoice data was then extracted into Excel files for processing. The data was cleaned and structured to ensure consistency and accuracy.

Next, data from multiple Excel sheets was merged, and record matching techniques were applied to identify correct entries and eliminate duplicates or mismatches.

Finally, the processed data was validated and organized into a structured format, making it ready for reporting and analysis. This workflow ensured efficient handling of financial data and reduced manual effort.

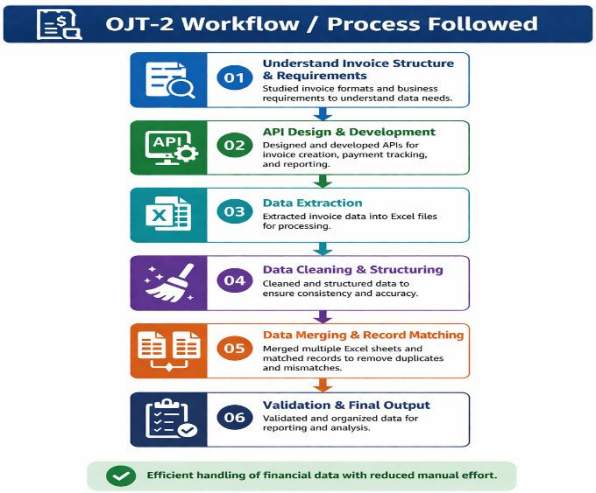


Fig 3.3 OJ2-Workflow

3.7 Conclusion :-

The OJT-2 phase provided valuable experience in invoice management systems and advanced API development. It helped in understanding how technical solutions are applied in real-world financial operations. The skills and knowledge gained during this phase improved my confidence and prepared me for handling complex tasks in a professional environment.

CHAPTER-04

USE CASE-01 AND USE CASE-02

4.1 USE CASE 1: GP-SMS AUTOMATION :-

4.1.1 Title :-

Automation of SMS Data Processing and Report Generation

4.1.2 Description:-

This use case focuses on automating the process of downloading SMS backup data, cleaning and transforming it, and generating structured Excel reports.

4.1.3 Steps Involved:-

- Download SMS data from source
- Clean phone numbers and remove unwanted data
- Convert data into Excel format
- Split and format date/time fields
- Merge data with GP master dataset
- Generate unique SMS records
- Create final output report

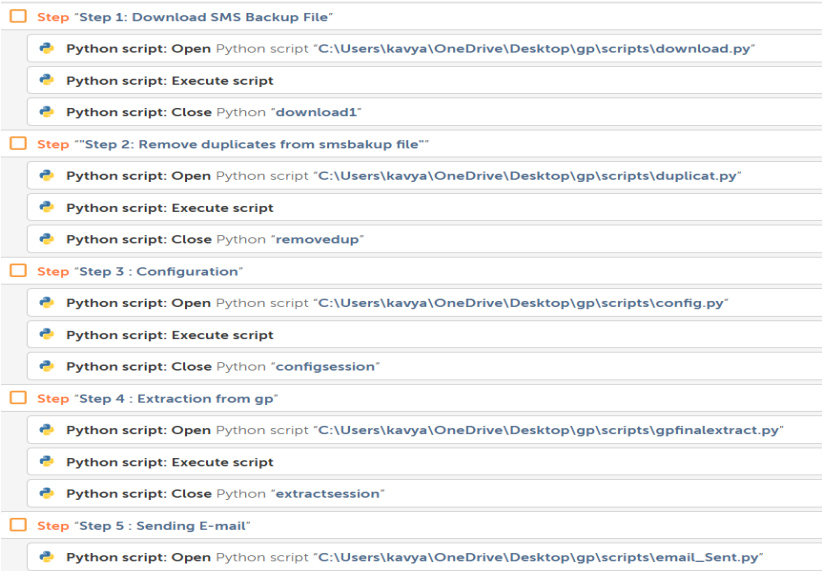


Fig 4.1 Project workflow

4.1.4 Tools Used :-

- Python (pandas)
- Microsoft Excel
- Automation Anywhere A360

4.1.5 Outcome :-

The system successfully automated SMS data processing, reducing manual effort and increasing efficiency and accuracy in report generation.

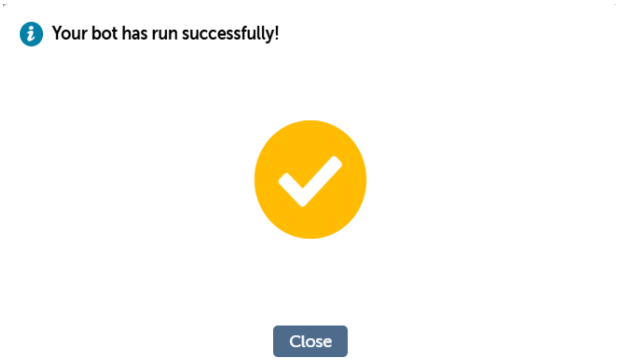


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Fig 4.2 Bot Completion and Processed Excel Report

4.2 USE CASE 2: DROUGHT PREDICTION SYSTEM

4.2.1 Title :-

Prediction of Drought Conditions using Machine Learning

4.2.2 Description :-

This use case focuses on predicting the likelihood of drought based on environmental parameters. The system takes user input through a web interface and provides prediction results using a trained machine learning model.

4.2.3 Steps Involved :-

- Collect dataset (rainfall, temperature, humidity, soil moisture)
- Preprocess and clean the data
- Train machine learning model
- Save trained model (model.pkl)
- Develop Flask web application
- Take user input through form
- Predict drought condition
- Display result to user

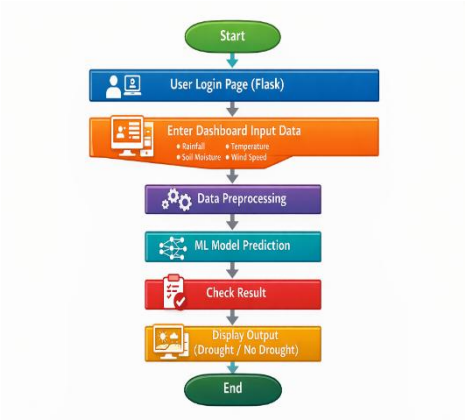

Fig 4.3 Project Workflow

4.2.4 Inputs :-

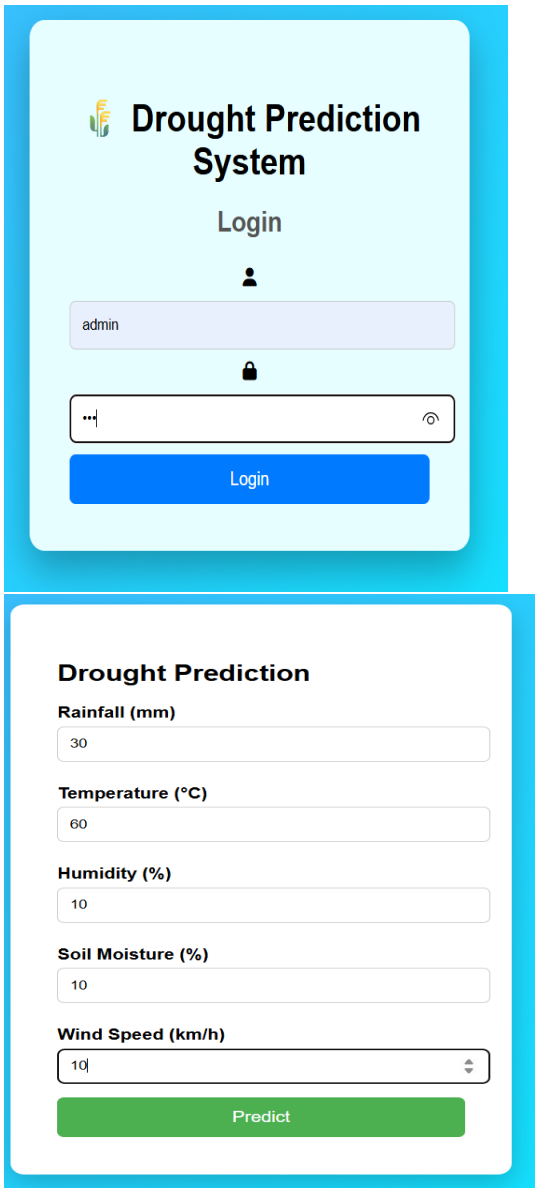

Fig 4.4 Login page and Input Dashboard Page

4.2.5 Tools Used :-

- Python
- Scikit-learn
- Flask
- Pandas
- HTML/CSS

4.2.6 Outcomes :-

The system successfully predicts drought conditions with good accuracy and provides a user-friendly interface for easy access. It helps in early decision-making and risk management

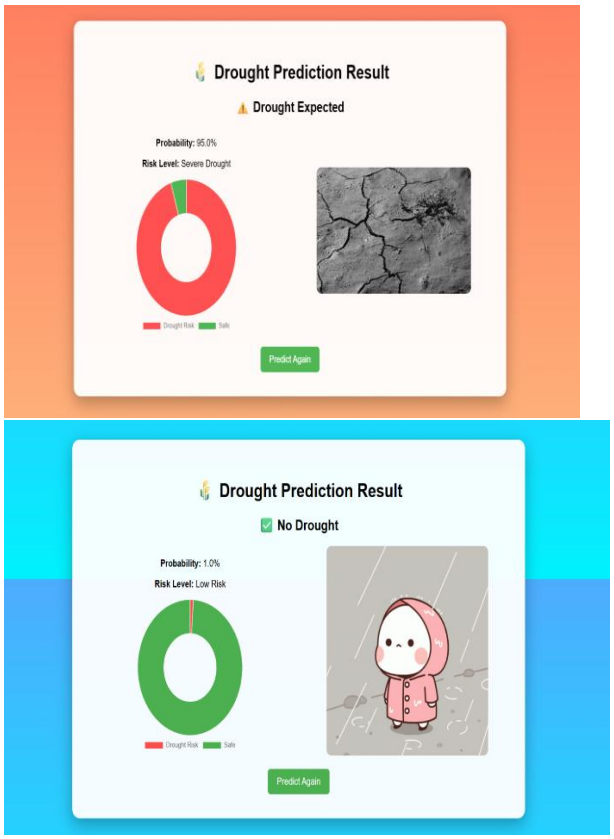

Fig 4.5 Output of the Project

4.2.7 Conclusion :-

The use cases developed during the internship demonstrated the practical application of technical concepts in real-world scenarios. The GP-SMS automation and invoice management systems helped in understanding how data processing, API development, and automation can be used to solve business problems efficiently.

These use cases improved accuracy, reduced manual effort, and enhanced system efficiency. They also provided hands-on experience in handling real-time data, building scalable solutions, and integrating different technologies.

Overall, Chapter 4 highlights the successful implementation of project-based learning and the ability to apply theoretical knowledge to practical industry applications.

Megha Bharathi M

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Objective

Motivated diploma student in Computer Science and Engineering with strong academic foundation and hands-on project experience. Skilled in DSA(python) and AI/ML. Eager to apply knowledge and contribute effectively in an entry-level position while continuing to learn and grow.

Education

- Diploma in branch of Computer Science and Engineering, SJ(Govt)Polytechnic,Bangalore.
  CGPA : **9.53** (6th SEMESTER RESULT IS AWAITED)
- Secondary(10th), Florence Public School, State – 2023
  Percentage : **92.16%**

Technical Skills

- **Software/tools** : GitHub, Jupyter Notebook
- **Programming** : Python
- **Other** : Data Structure with python

Academic Projects

1. **Project on GP-SMS – 4 weeks**
   Breif : Worked on automation
   Tools/technologies : Python,VScode
2. **Project on Agriculture Drought Prediction – 4 weeks**
   Breif : Worked on Randome Forest classifier , Prediction
   Tools/technologies : Python,Pandas

Certifications

- **Java Training –** IIT Bombay
- **Git Training –** IIT Bombay
- **Effective Speaking and listening Skills –** Wadhwani foundation
- **AI/ML-** at Netlabs Global IT services Pvt Ltd

Personal Details

- **Date of Birth –** 10/09/2007
- **Languages known** - English, Hindi, Kannada,Telugu
- **Address** – Whitefield, Bangalore-560067, Karnataka

PHOTO GALLERY :-

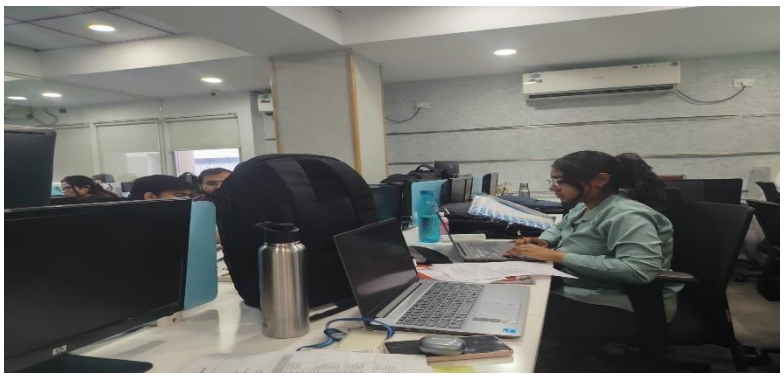

Fig 6.1 Student is working

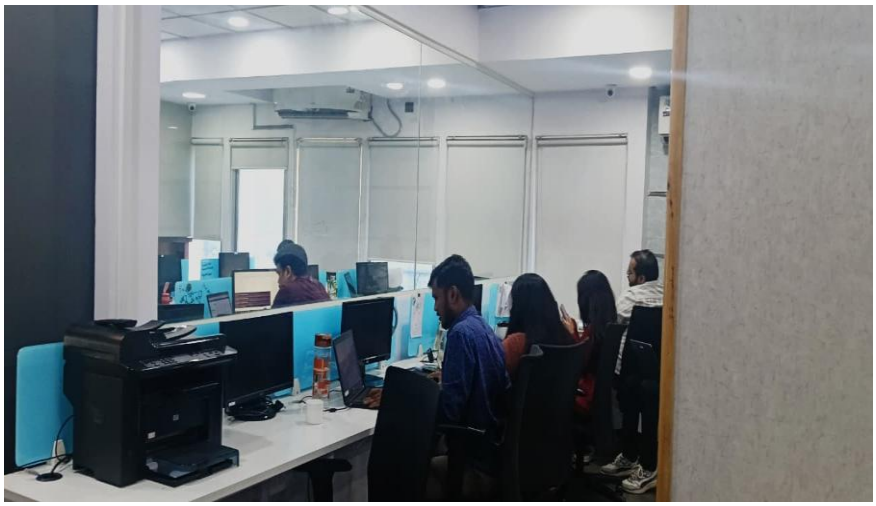

Fig 6.2 Employees are working

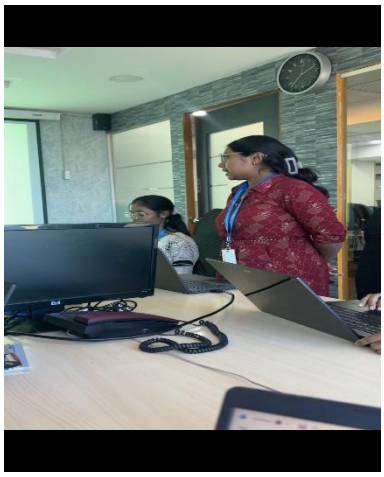

Fig 6.3 Student is giving presentation in Conference room

APPENDICES

Appendix A: GP-SMS Automation :-

This system is designed to automatically process SMS data and convert it into organized Excel reports.

Instead of doing manual work, the system:

- Collects SMS data
- Cleans unnecessary information
- Organizes it into proper format
- Generates a final report

In simple terms:
It works like a smart tool that takes messy data and turns it into a clean and useful report automatically.

Appendix B: API Development :-

APIs are like a bridge that allows different systems to communicate with each other.

In this project:

- APIs were used to send and receive data
- Different types of APIs were created for various tasks
- Security methods were used to protect data

In simple terms:
APIs act like a messenger that safely transfers information between systems.

Appendix C: Authentication and Security :-

Different authentication methods were used to ensure that only authorized users can access the system:

- Token-based authentication
- Session-based authentication
- API key authentication

In simple terms:
These methods work like passwords or ID cards that control who can use the system.

Appendix D: Invoice Management System :-

The system manages invoices, payments, and financial records using APIs and data processing techniques.

It helps to:

- Track payments
- Maintain records
- Generate reports

In simple terms:
It is like a digital system that keeps track of bills and payments automatically.

Appendix E: Excel Data Processing :-

Excel was used to:

- Store data
- Merge different datasets
- Match records
- Generate reports

GITHUB LINK OF THE ENTIRE PROJECT

LINK :

https://github.com/Megha456-afk/Internship_report.git